IE UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

Machine Learning & Communication Skills – Sentence Outline

FIELD: How Much Does the Field of Study Shape University GPA Rankings?

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FIELD: Field Influence on Evaluation Levels across Degrees.

1 Introduction

1. [Adam: universities often allocate scarce places by ranking students from all degrees on one GPA list, which assumes a grade means the same in every degree.]
   a. [Adam: grading standards differ between departments of the same university [1, 2].]
   b. [Adam: the gaps are large and repeat across 33 US public universities [3].]
2. [Adam: research question, stated as a claim: we measure how much of a student's position in a pooled GPA ranking depends on the field of study, and how ranking within degree changes who is selected.]
   a. [Adam: what the project measures (grading differences between fields) and what it does not claim (intrinsic difficulty of a degree).]
   b. [Adam: one sentence that previews the main finding and the structure of the report.]

2 Data

3. [Pablo: three complementary sources answer different parts of the question, and they are analysed separately because they share no key.]
   a. [Pablo: Spanish SIIU table: 12,125 university-field-year means, 89 universities, 29 fields, 2015-16 to 2023-24, with quartiles [4]; limits: aggregated, graduates only.]
   b. [Pablo: Portuguese student records: 3,703 students in 15 degrees with standardised entry grades [5].]
   c. [Adam: US data: what the 33-university dataset adds and how departments were mapped to fields [3].]

3 Methodology

4. [Pablo: the analysis separates shared field effects from university-specific ones and tests them on data the models have not seen.]
   a. [Pablo: variance decomposition with fixed effects, re-estimated with a mixed-effects model with random university and university-by-field effects.]
   b. [Pablo: temporal validation (train 2015–2021, test 2022–2023) with bootstrap intervals over universities.]
   c. [Pablo: confounders (standardised admission grade, size, dropout) and why dropout is treated separately.]
5. [Pablo/Nacho: the selection simulation and the comparison of ranking rules translate grade gaps into places won or lost.]
   a. [Pablo: the simulation approximates within-field grades from mean and quartiles, and its normal assumption was checked against reported medians and maxima.]

b. [Nacho: four ranking rules are compared by ranking on first-semester grades and evaluating them on second-semester grades.]

4 Results and Discussion

6. [Nacho: most of the variation in grades within a university is explained by the field of study, and part of it is specific to each university.]
   a. [Nacho: 65% shared field effect, 28% university-specific, 7% noise; the mixed model gives 61 / 29 / 11%.]
   b. [Nacho: the university-specific part is stable over time (r = 0.84, 95% CI 0.84–0.85).]
   c. [Nacho: past field profiles predict later years best (test RMSE 0.137 vs 0.226 for the shared effect alone).]
7. [Nacho: intake differences do not explain the field gaps.]
   a. [Nacho: admission grade and size leave field effects almost unchanged (SD 0.31, correlation 0.99); in the Portuguese records the gap only falls from 1.51 to 1.30.]
   b. [Nacho: dropout explains part of the spread, but low grades may themselves cause dropout.]
8. [Adam: the US replication result: whether the ranking of fields is similar in both countries (correlation of field effects).]
   a. [Adam: main supporting number.]
   b. [Adam: main difference or limitation of the comparison.]
9. [Nacho: a single raw-GPA ranking gives very unequal access across fields, and the result is robust to the distributional assumption.]
   a. [Nacho: 41% of Nursing graduates vs 6% in Business Administration reach a top-20% cut (median university-year, with interquartile ranges).]
   b. [Pablo: replacing the normal with a truncated normal changes shares by 0.04 percentage points on average and no field changes rank.]
10. Ranking students within their own degree would make access to a top-20% selection much more equal across degrees, with almost no change in the later performance of the selected students.
    a. The selected students obtain 14.21 out of 20 in the second semester under raw GPA and 14.08 under within-degree percentiles, but the 95% confidence intervals overlap, so the difference is too small to be conclusive.
    b. Under raw GPA some degrees obtain many more places than others, whereas with within-degree percentiles almost all degrees obtain a similar share, as the spread of selection rates across degrees falls from 0.090 to 0.006.
    c. The higher rank stability of raw GPA (0.66 against 0.62) does not make it a better rule, because it partly comes from the gap between degrees repeating every semester, which shows that the inequality is persistent rather than occasional.
11. [Nacho: the main limitations and what they mean for the conclusions (aggregated data, graduates only, public universities for admission grades).]

5 Conclusions and Future Work

12. [Nacho: synthesis: pooled GPA rankings sort students largely by field, and within-degree normalisation reduces this at little cost.]
13. [Nacho: future work, e.g. individual-level data from more institutions to separate grading practice from student ability.]

References

[1] R. Sabot and J. Wakeman-Linn, "Grade inflation and course choice," *Journal of Economic Perspectives*, vol. 5, no. 1, pp. 159–170, 1991, doi:10.1257/jep.5.1.159.

[2] A. C. Achen and P. N. Courant, "What are grades made of?," *Journal of Economic Perspectives*, vol. 23, no. 3, pp. 77–92, 2009, doi:10.1257/jep.23.3.77.

[3] H. Jeong, G. M. Weiss, A. Leung and D. D. Leeds, "The construction and analysis of course grades across public universities," in *Proc. 17th Int. Conf. Educational Data Mining*, 2024, pp. 643–648, doi:10.5281/zenodo.12729906.

[4] Ministerio de Ciencia, Innovación y Universidades, "Media, cuartiles y máximo de la nota del expediente (estudiantes egresados) por ámbito de estudio y sexo," SIIU, 2025. [Online]. Available: https://estadisticas.ciencia.gob.es

[5] V. Realinho, M. Vieira Martins, J. Machado and L. Baptista, "Predict students' dropout and academic success," UCI Machine Learning Repository, 2021, doi:10.24432/C5MC89.

Declaration of Contributions and AI Use

[All: one entry per member describing what each actually did (analysis, code, sentences, reviews), and every use of AI tools with tool, purpose, output used and verification.]